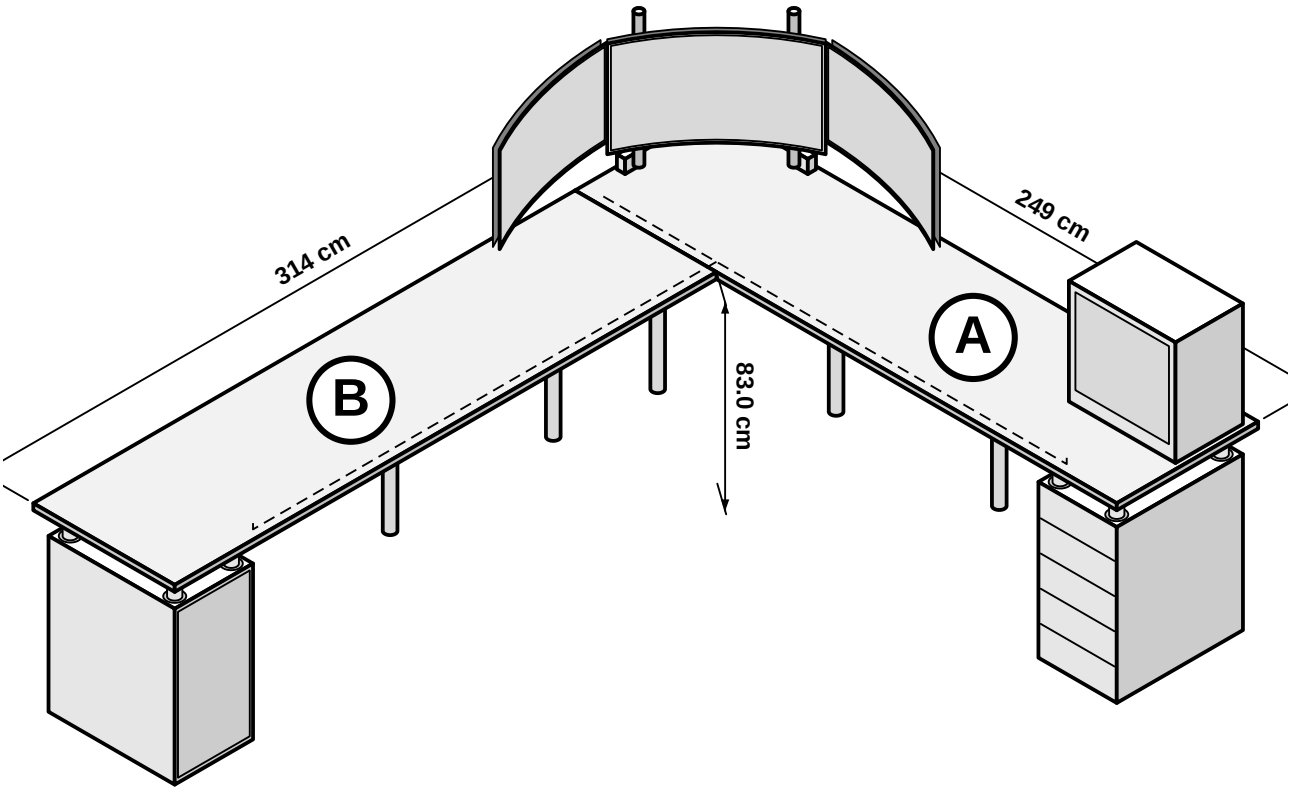
L-Shaped Corner Desk

Assembly manual



Footprint	249 × 314 cm
Surface height	83.0 cm
Tops	A along the back wall, B along the side wall (two IKEA EKBACKEN, each 249 × 65 cm, 28 mm thick)
Floor needed	252 × 317 cm, including the 3 cm wall gaps

Tape-measure sizes in cm. The rails are hidden under the front edges and are drawn dashed.

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A · Finished assembly

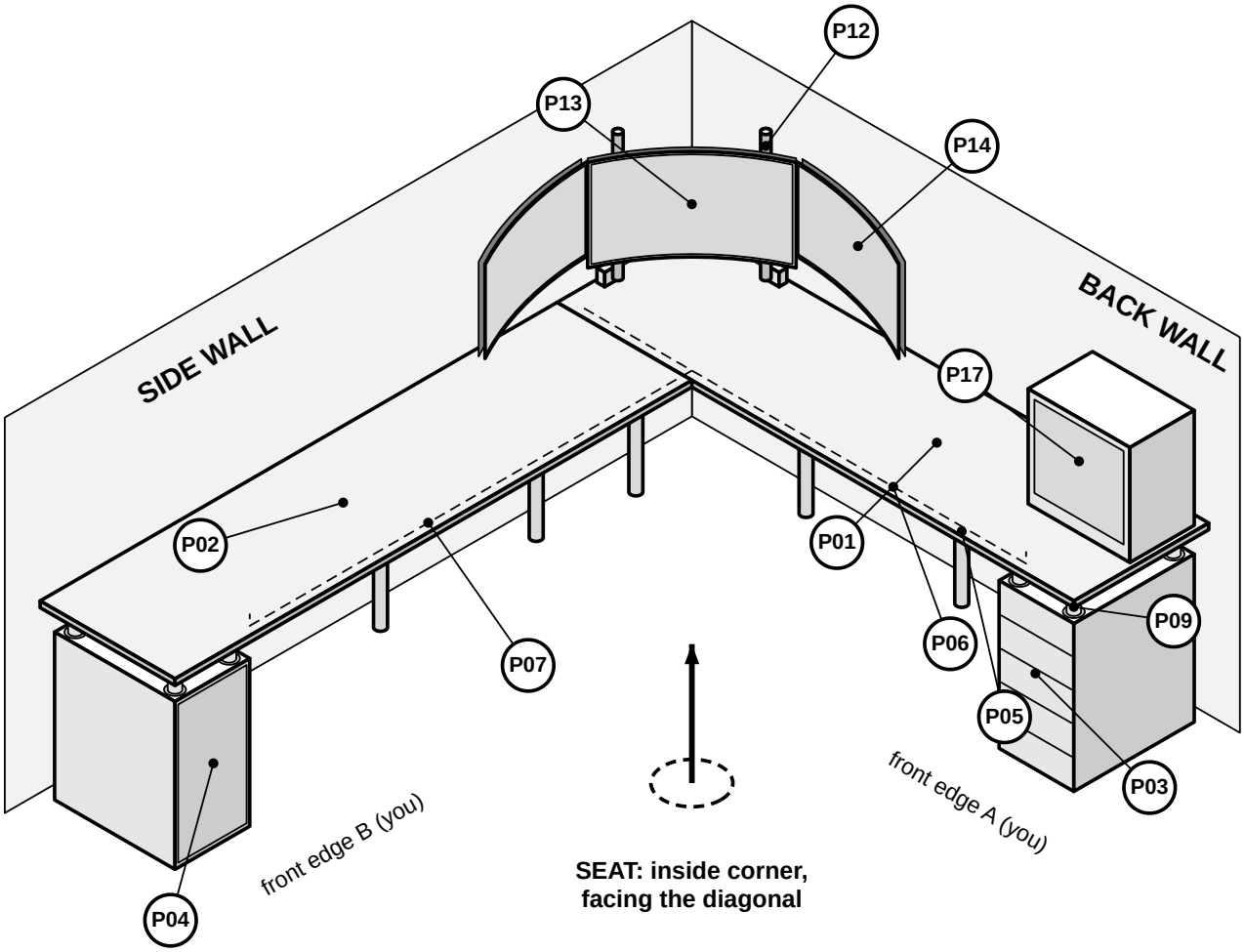


Fig. A-1 The finished desk seen from the room. Dashed lines under the front edges: the two steel rails, hidden by the tops. Left and right are as you sit at the inside corner facing the diagonal: top A is on your right, top B on your left.

No.	Part	No.	Part
P01	Top A, EKBACKEN 249 × 65 cm, 28 mm thick	**P07**	Rail B, steel angle 213 cm
P02	Top B, EKBACKEN 249 × 65 cm, 28 mm thick	**P09**	Round adjustable foot 4 in x 8
P03	ALEX 1, drawer unit 36 × 58 × 70 cm	**P12**	VIVO STAND-V300CU
P04	ALEX 2, storage unit with door 36 × 58 × 70 cm	**P13**	Samsung Odyssey G5 32 in
P05	OLOV leg x 6, 105 mm plate	**P14**	Acer XZ320QU x 2
P06	Rail A, steel angle 213 cm	**P17**	HYTE Y60 on printed pedestal

What this desk is

Two IKEA kitchen tops make an L. Top A lies along the back wall and keeps the corner square; top B lies along the side wall and its end butts against top A's front edge. Each top stands 3 cm off its wall, with its square, unfinished long edge toward the wall. The far end of each top rests on an ALEX cabinet through four adjustable feet: the feet are screwed to the top, and their pads drop into printed rings on the cabinet, so the desk cannot slide on the cabinets and the two heavy boxes hold the whole frame square. Six OLOV legs stand along the two wall edges, none at the front, so the space under the inside corner is open for your knees.

That leaves the two front edges with nothing under them for more than two metres, so each one is carried by a steel angle rail screwed to the underside 12 cm back from the front edge. Under the inside corner the two rails are bolted together with two gusseted steel fittings, so the corner of top B hangs from rail A. The rails never touch a cabinet: their ends load the top, and the top loads the nearest feet. The monitor stand clamps to top A's two wall edges, 40 cm from the corner; the surface is 83.0 cm above the floor.

A · Plan of the finished desk

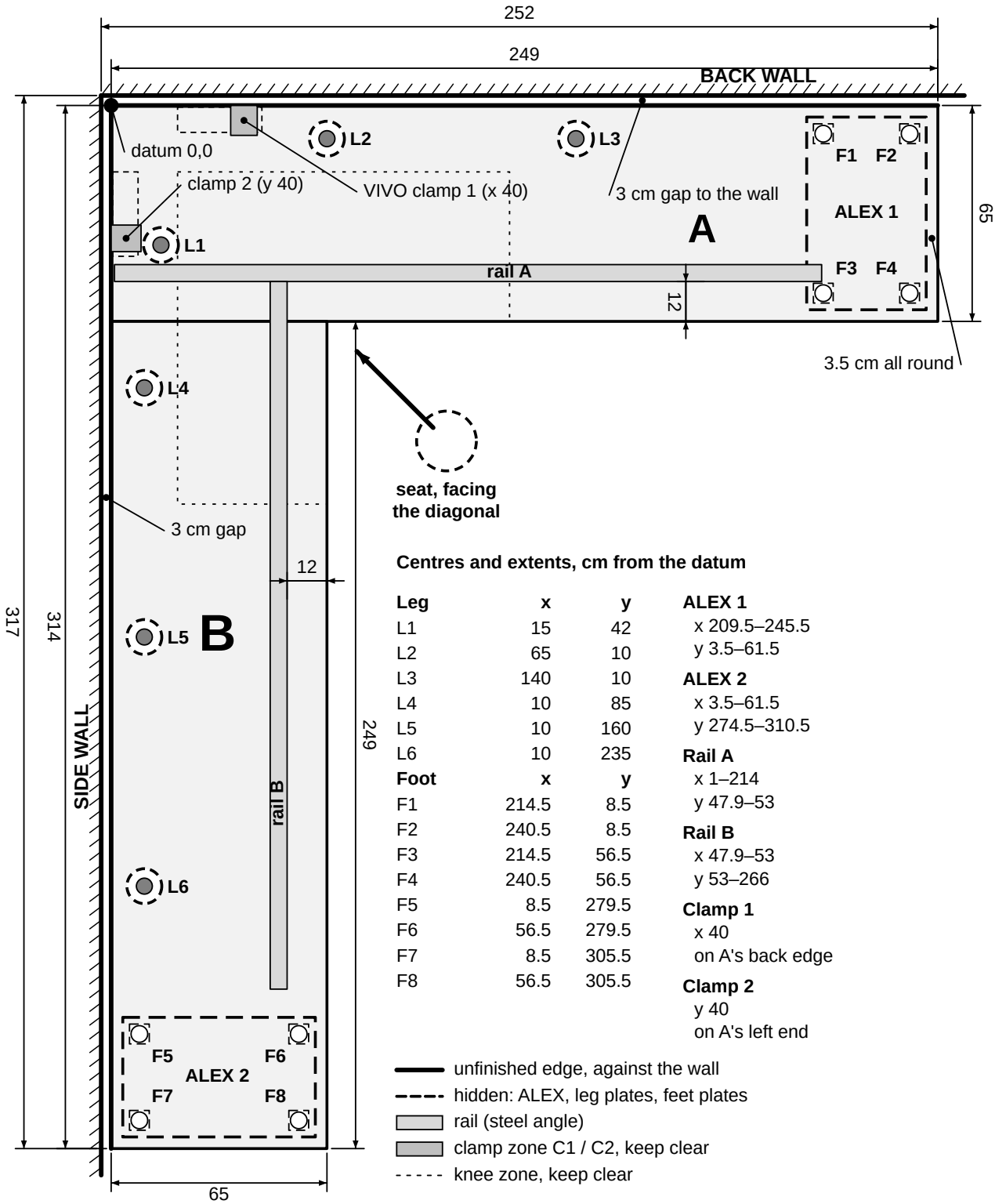


Fig. A-2 Top-down plan, seen from above. Datum: top A's back-left corner (0, 0); x runs along the back wall, y along the side wall, all in cm. Coordinates SOURCE-DEFINED (docs/build-manual.md). ALEX 1 drawers and ALEX 2 door face you.

Height above the floor	cm	Confidence
Top surface	83.0	SOURCE-DEFINED
Underside of the tops (OLOV setting)	80.2	SOURCE-DEFINED
Rail lower edge	75.1	SOURCE-DEFINED
ALEX top (the feet stand on it)	70.0	ASSUMPTION / FIELD CHECK measure your units

G · Legs and cabinet support

Six OLOV legs and eight feet plates go on the tops while they lie face down on a blanket; the cabinets stand by the walls; then each top is stood on its bare cabinet once so its pads can mark where the eight locating rings go.

No.	Part	No.	Hardware	Qty
P01 **P02**	Top A and top B, EKBACKEN 249 × 65 cm, 28 mm thick	**H01**	M4 x 16 mm machine screw, IKEA 108443 (with the OLOV legs)	30
P03 **P04**	ALEX 1, drawer unit; ALEX 2, storage unit with door (36 × 58 × 70 cm)	**H02**	#8 x 5/8 in pan head screw (feet plates, 4 each)	32
P05	OLOV leg x 6, 105 mm plate	**H06**	#6 x 5/8 in flat head (rings, 3 each)	24
P09 **P10**	Round adjustable foot 4 in x 8; printed locating ring x 8	**P18** printed tools: leg jig, feet corner jig, edge T-square, ALEX corner template, drill guide (tower + pucks), foot height gauge, wall gauge		

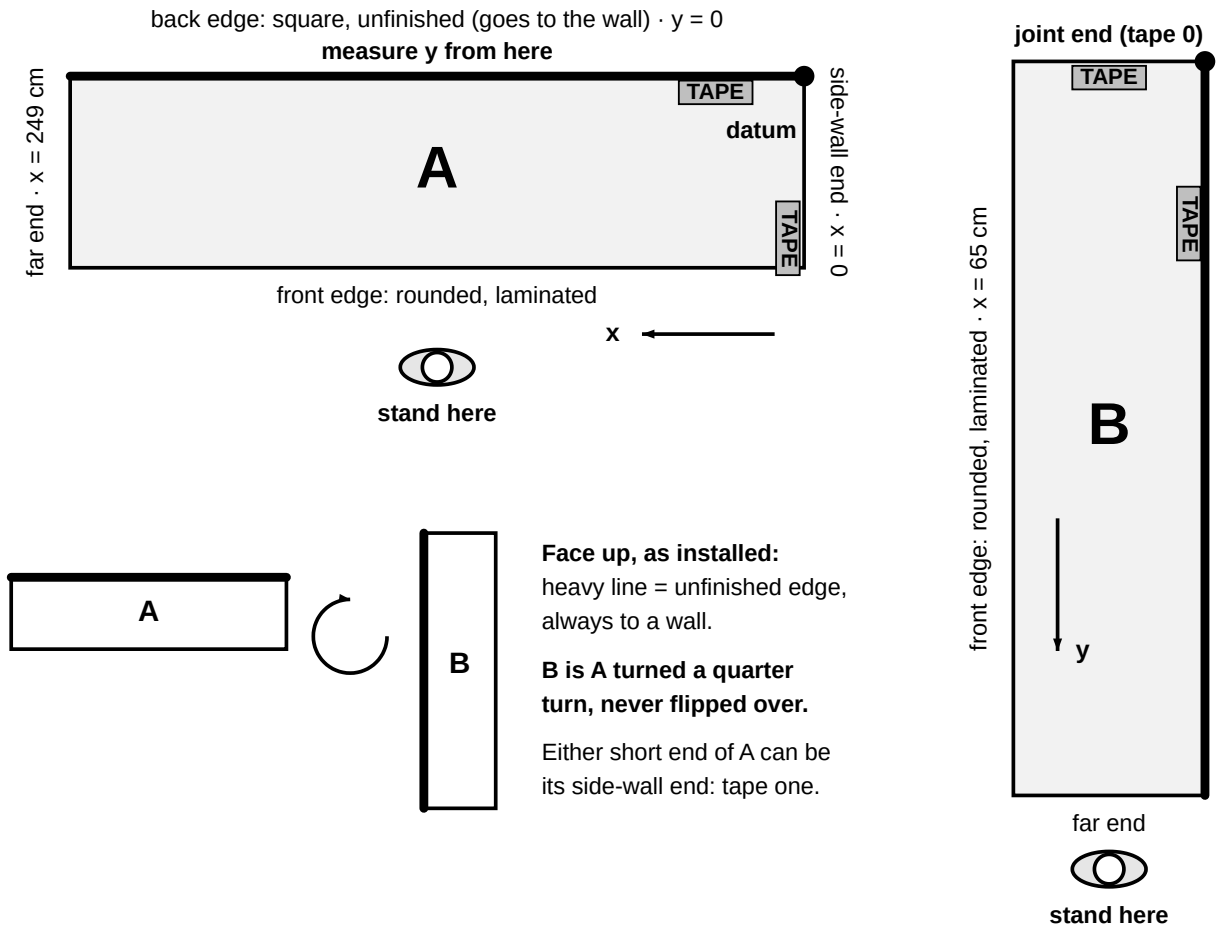


Fig. G-1 How each top lies while you mark it: face down, laminate on the blanket. Heavy line: the square, unfinished long edge, which always goes to a wall (y = 0 on A, x = 0 on B). Tape the unfinished edge and the end you measure from, and stand where the person symbol is; the drawings in this chapter then match what you see, with x growing to the left. Datum: top A's back-left corner (0, 0) as installed.

Decide the desk height first

The legs are pre-set in step 1, so choose now. The desk is built at 83.0 cm as planned. Optional printed ALEX feet (2, 4 or 6 cm tall, ⌀56 mm, `parts/accessories/alex-foot.scad`) raise everything by their height; nothing on the desk moves. **SOURCE-DEFINED**

ALEX feet	Underside of the tops (cm)	Top surface (cm)	Every OLOV set to (cm)	Rail lower edge (cm)
none (as planned)	80.2	83.0	80.2	75.1
2 cm	82.2	85.0	82.2	77.1
4 cm	84.2	87.0	84.2	79.1
6 cm	86.2	89.0	86.2 (OLOV max 90)	81.1

G · Underside support layout

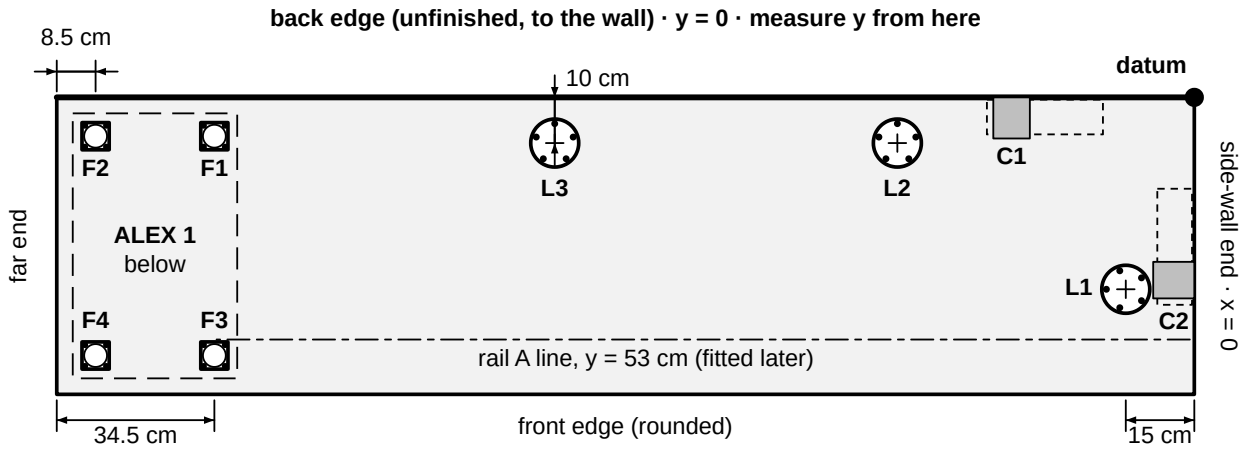


Fig. G-2 Top A face down, back edge away from you: x grows to the left from the side-wall end. Circles: OLOV plates with their five holes, one pointing at the wall (L1's pattern is turned). Squares: feet plates. Grey: clamp zones C1, C2, keep clear; thin dashed: the clamp pad outlines; dash-dot: rail A's line. Datum (0, 0) at the top-right corner of this view. Drawing labels in cm.

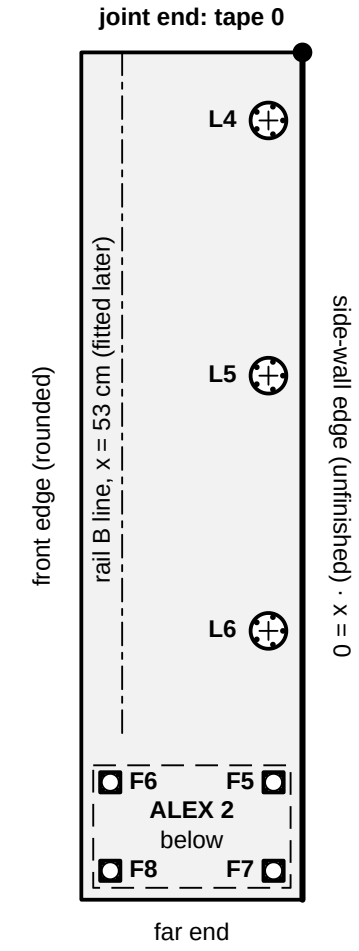


Fig. G-3 Top B face down, joint end away from you. Hook the tape on the joint end; the table gives the reading for each mark.

Mark	x	y	Tape reading
L1	15	42	42 from A's back edge
L2	65	10	65 from the side-wall end
L3	140	10	140 from the side-wall end
L4	10	85	20 from B's joint end
L5	10	160	95 from the joint end
L6	10	235	170 from the joint end
F1	214.5	8.5	214.5 from the side-wall end
F2	240.5	8.5	corner jig, far back corner
F3	214.5	56.5	214.5 from the side-wall end
F4	240.5	56.5	corner jig, far front corner
F5	8.5	279.5	214.5 from the joint end
F6	56.5	279.5	214.5 from the joint end
F7	8.5	305.5	corner jig, far wall corner
F8	56.5	305.5	corner jig, far front corner

x, y and tape readings in cm; x, y from the datum **SOURCE-DEFINED**. Feet are 5 cm in from each ALEX corner, which is 8.5 cm in from the top's edges. F1 F3 F5 F6 are also 34.5 cm from the far end.

Leg plates are ⌀105 mm and feet plates 59.5 mm square. L1's plate ends short of rail A and clear of the clamp 2 jaw; feet plates F3 and F6 stay 5 mm clear of the rails.

> Nothing here goes in the clamp zones C1 (36–44 cm from the side-wall end, the 9 cm next to the back edge) or C2 (36–44 cm from the back edge, the 9 cm next to the side-wall end).

G · Step 1: legs

> **MEASURE ON HARDWARE** MEASURE ON HARDWARE
> Calipers on both tops before the first hole: expect 28 mm. If a top reads 27 mm, use screws 1 mm shorter. No screw may go deeper than 21 mm into a top.

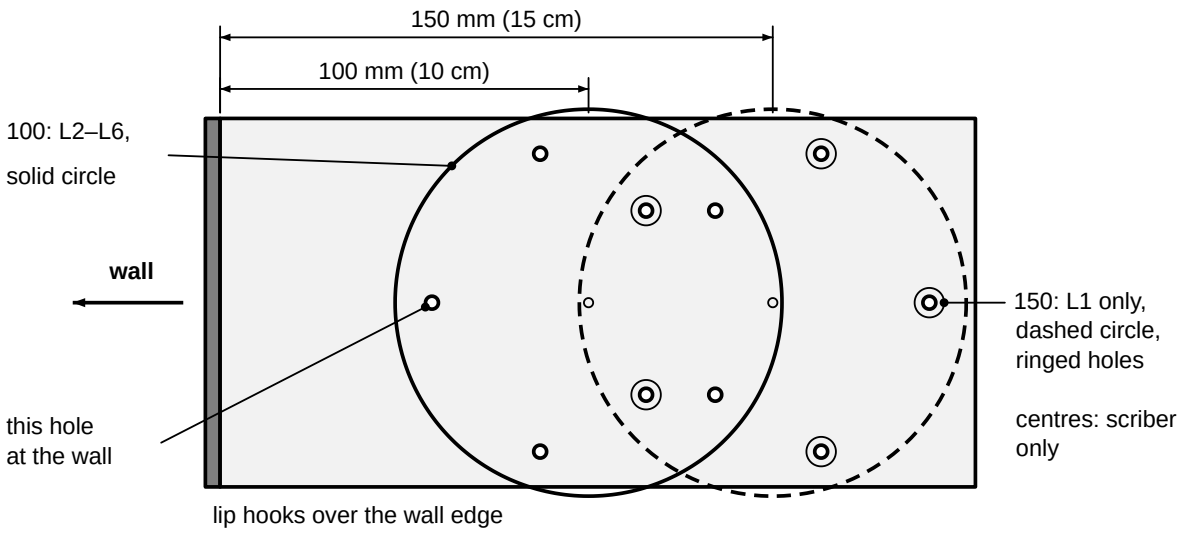


Fig. G-4 Leg jig (P18), seen from above in use, lip hooked down over the wall edge. The 100 pattern (10 cm from the edge; solid plate circle, plain holes) is for L2 to L6 and puts one hole at the wall; the 150 pattern (15 cm; dashed circle, ringed holes) is for L1 only.

Leg	Top	Hook the lip over	Tape station (cm)	Pattern
L1	A	side-wall end (x = 0)	42 from the back edge	150, ringed holes
L2 L3	A	back edge (y = 0)	65 (L2) and 140 (L3) from the side-wall end	100
L4 L5 L6	B	side-wall edge (x = 0)	20 (L4), 95 (L5), 170 (L6) from the joint end	100

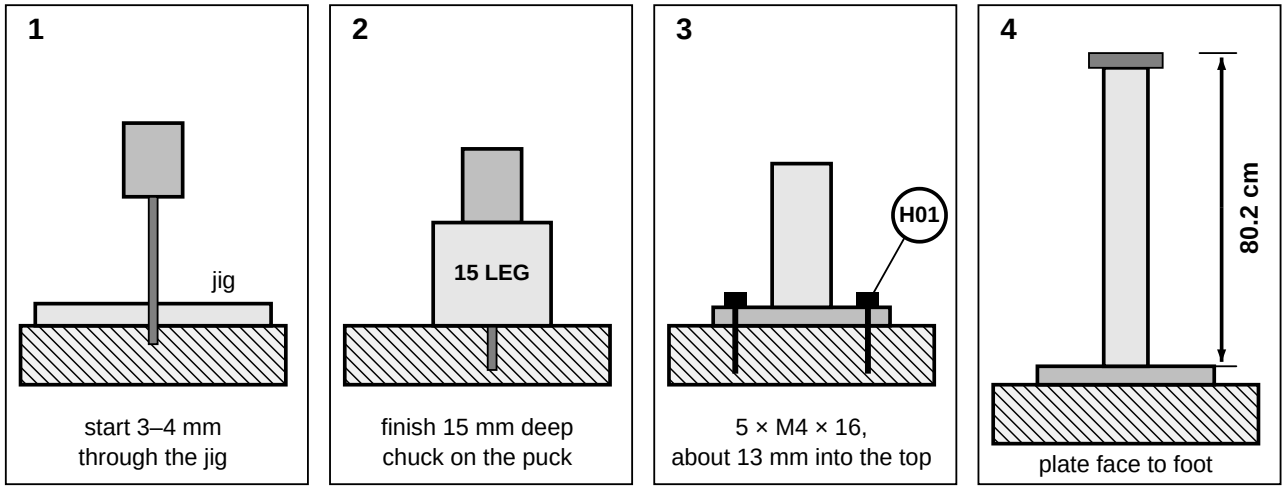


Fig. G-5 One leg, four operations (steps 1a–1d below). The drill guide's SET tower sets the 1/8 in bit first.

1

a Jig on the station, start all five pilots 3–4 mm through its holes with the 1/8 in bit. **b** Lift the jig; finish each pilot 15 mm deep with the **15 LEG** puck. **c** Screw the OLOV plate on with five **H01** M4 × 16 screws (about 13 mm into the top). **d** Pre-set the leg to **80.2 cm** from plate face to foot (or the table's number on page G-1 if you fit ALEX feet). Six legs, 30 screws.

OLOV plate ⌀105 mm, five ⌀5.5 mm holes on an ⌀85 mm circle, neighbouring holes 50 mm apart; leg range 60–90 cm. VERIFIED
105 mm plate measured 22 Oct 2026; hole pattern from a shared OLOV template, printed and test-fitted.

G · Step 2: feet plates

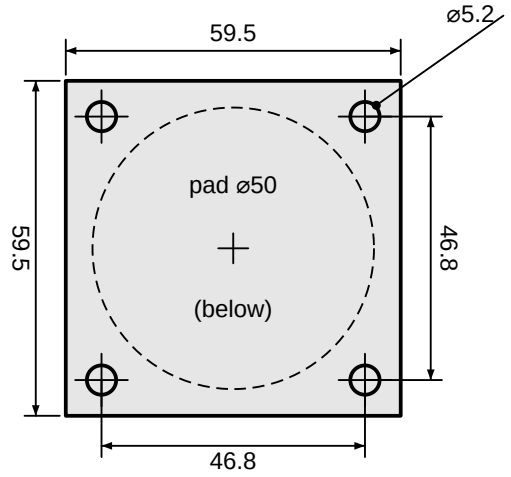
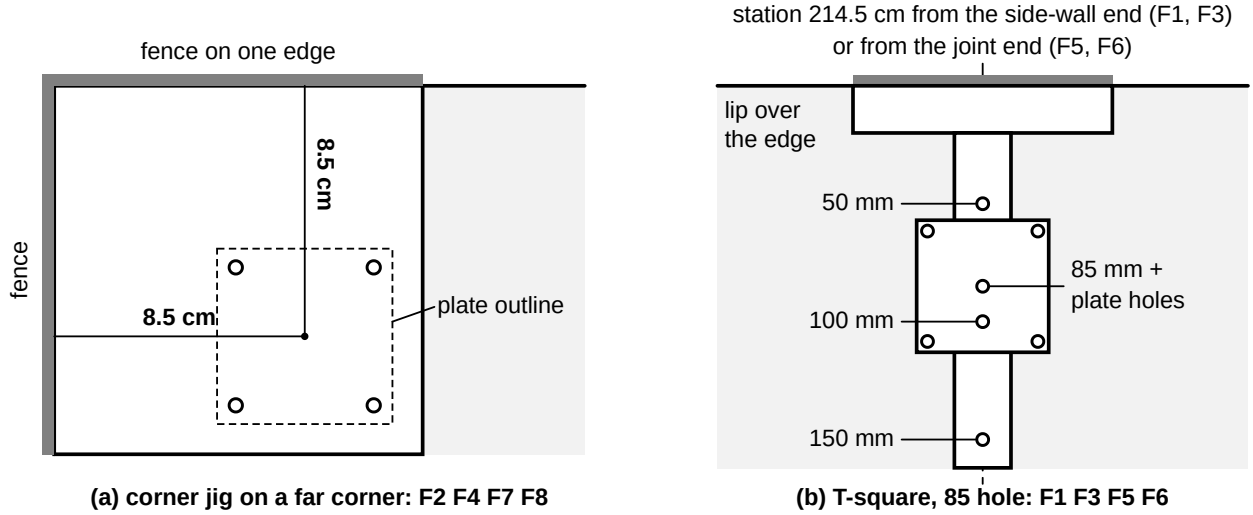


Fig. G-6 Feet plate (P09) from the screw side, mm. VERIFIED on the purchased plate, 23 Oct 2026: 41.6 mm between two neighbouring holes (inside jaws), 52.0 mm across them (outside jaws).

Feet	Tool	Placement
F2 **F4**	feet corner jig	top A's two far corners, fences on both edges
F1 **F3**	T-square, 85 hole	lip over A's back edge (F1) or front edge (F3), station 214.5 cm from the side-wall end
F7 **F8**	feet corner jig	top B's two far corners
F5 **F6**	T-square, 85 hole	lip over B's side-wall edge (F5) or front edge (F6), station 214.5 cm from the joint end

ASSUMPTION / FIELD CHECK
ASSUMPTION / FIELD CHECK
The feet's load rating is unknown; the design wants 50 kg or more each.



(a) corner jig on a far corner: F2 F4 F7 F8

(b) T-square, 85 hole: F1 F3 F5 F6

Fig. G-7 Both tools put the foot's four plate holes (1/8 in, on the 46.8 mm square) square to the edges in one placement, seen face down. Centre holes are for a scriber only. T-square offsets are in mm, as engraved on it.

2 Start the four pilots 3–4 mm through the tool's plate holes with the 1/8 in bit, lift the tool, finish 15 mm deep with the **15 LEG** puck. Screw the plate on with four **H02** #8 × 5/8 pan heads (14 mm into the top), pad pointing away from the top. Eight plates, 32 screws.

3 Wind each pad out until the printed foot height gauge (102.0 mm inside) just slides over the foot from the side with no rock. All eight then match without a ruler.

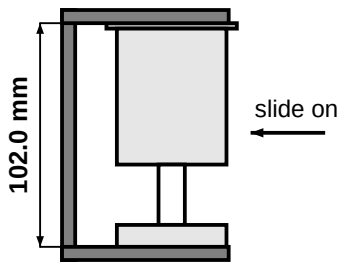


Fig. G-8 Foot height gauge.

G · Step 4: place the ALEX units

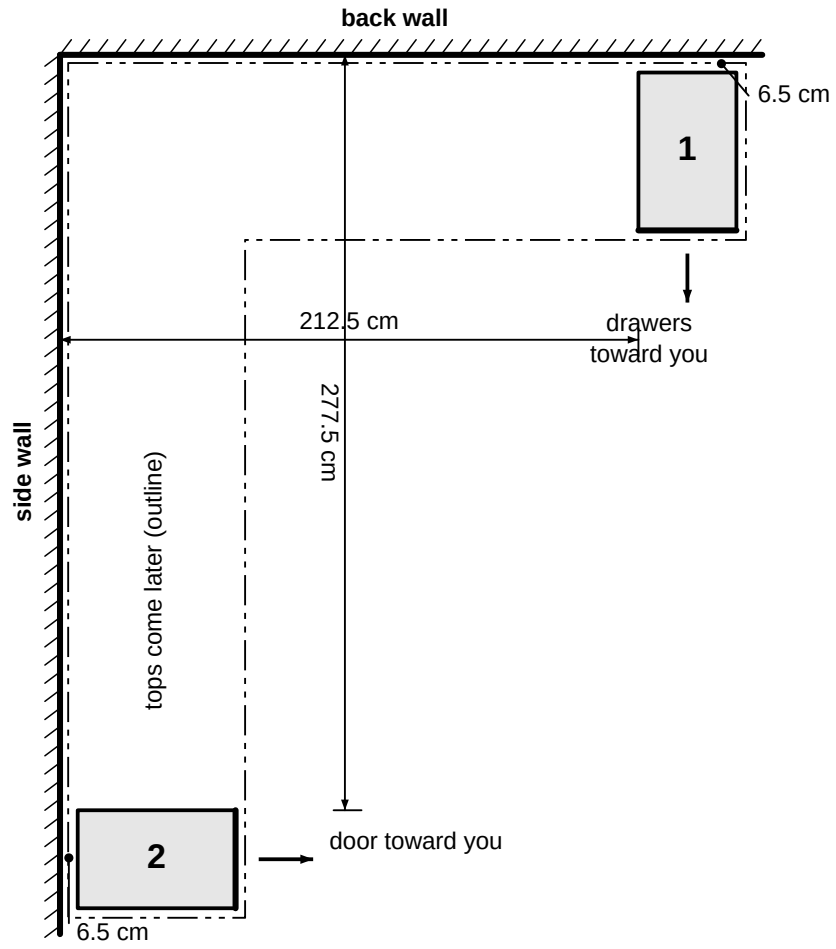


Fig. G-9 Plan from above. The tops are not up yet, so the cabinets are set from the walls (cm). Heavy line: the drawer fronts of ALEX 1 face you (+y); the door of ALEX 2 faces you (+x).

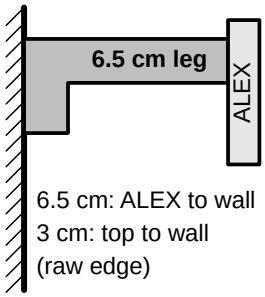


Fig. G-10 Wall gauge (P18): the 6.5 cm leg just fits between the wall and a cabinet; the 3 cm leg checks a top's raw edge.

4

Stand **ALEX 1** (P03) with its back 6.5 cm off the back wall and its near side 212.5 cm from the side wall, drawers toward you. Stand **ALEX 2** (P04) with its side 6.5 cm off the side wall and its near end 277.5 cm from the back wall, door toward you. Painter's tape on the floor round each one saves guessing. **No rings yet:** they go on in step 5, traced from the pads.

Once the tops are on, 3.5 cm of top shows all round each cabinet and the tops stand 3 cm off the walls. That is only roughly right at this point; step 5 sets it exactly by pushing the cabinet under the standing top.

Optional ALEX feet (decided on page G-1): fit them now, before the cabinets stand. Unscrew the stock feet, lay the unit on its back, hook the corner template over each bottom corner and use its inner three holes (r 14.5 mm). Pilot 5/64 in, 12 mm deep with the 12 ALEX puck; three #6 × 5/8 flat heads per foot down the wells, four feet per unit, all the same height.

! **ASSUMPTION / FIELD CHECK** ASSUMPTION / FIELD CHECK
The drawings take an ALEX as 70 cm tall, which is what makes the feet 102 mm between the ALEX top and the underside at 80.2 cm. Measure your units, with and without their stock feet, before you wind the feet in step 3.

G · Step 5: rings, by tracing the pads

A 50 mm pad in a 52 mm ring leaves about 1 mm each side, closer than a cabinet can be placed with a tape. So the pads mark their own rings: the top stands on the bare cabinet first. Two people.

> **MEASURE ON HARDWARE** MEASURE ON HARDWARE
> **Pad diameter:** calipers on one pad before you print the rings. The ring is bored 52 mm for a 50 mm pad. If yours differs, change `pad_d` in `parts/structure/foot-ring.scad` and reprint.

VERIFIED ALEX top panel 15.9 mm (5/8 in), measured 23 Oct 2026: the #6 × 5/8 screws reach 13.4 mm into it and leave 2.5 mm; the 12 mm pilot leaves 3.9 mm.

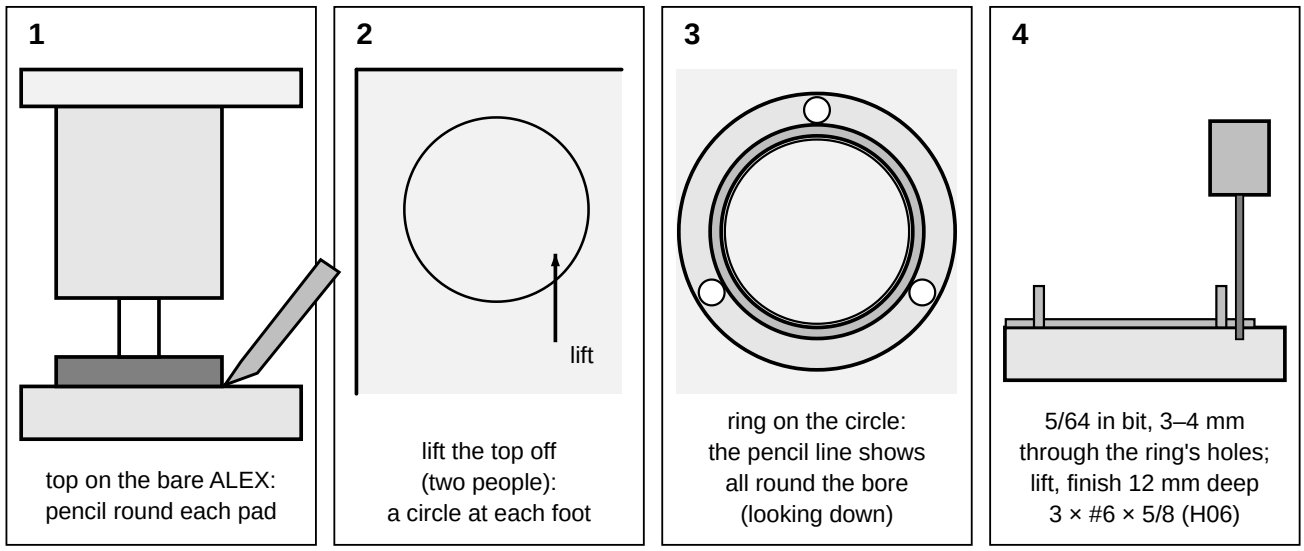


Fig. G-11 Fitting one ring (steps 5a–5d). Panels 1 and 4 are sections, mm; panels 2 and 3 are seen from above.

5

a Stand the top on its legs with its four pads on the **bare** ALEX top. Push the cabinet, never the top, until 3.5 cm of top shows all round it and the wall gauge's 6.5 cm leg just fits between the wall and the cabinet. Pencil round each pad. **b** Lift the top off onto the blanket (two people). **c** Set a ring on each circle and look down through its open bottom until the pencil line shows all the way round the bore. **d** Start the three pilots 3–4 mm through the ring's own holes with the 5/64 in bit, lift the ring, finish 12 mm deep with the **12 ALEX** puck, put the ring back and drive three **H06** #6 × 5/8 flat heads. Then stand the top up again: the pads drop into the rings. From now on move the cabinet, not the top.

G · The ring, and the check

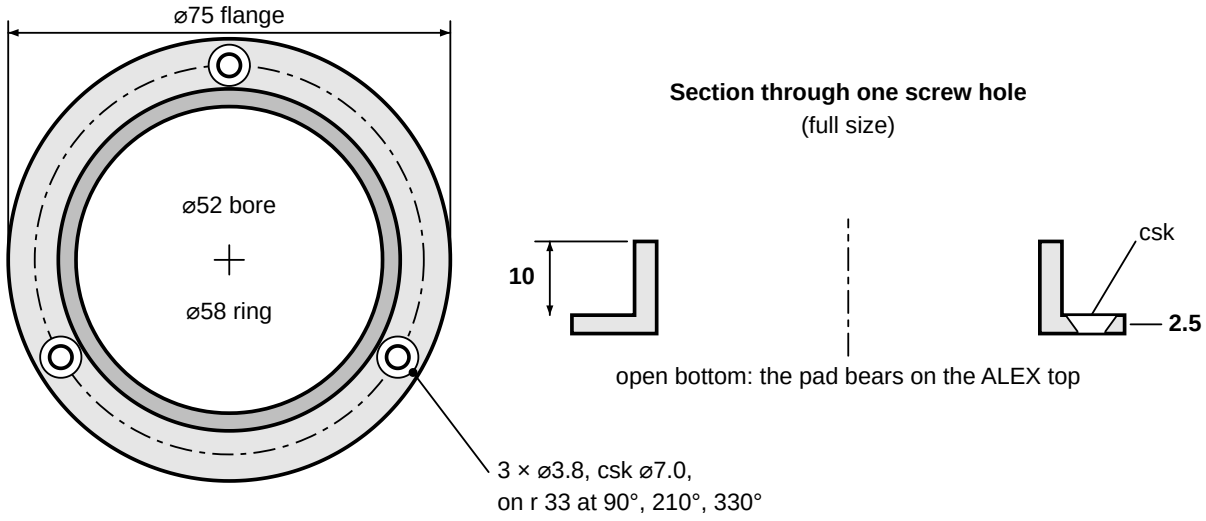


Fig. G-12 Printed locating ring (P10), from `parts/structure/foot-ring.scad`. VERIFIED from its own geometry file, mm: bore ø52, ring ø58 (3 wall) and 10 tall on a ø75 × 2.5 flange, three ø3.8 holes countersunk ø7.0 on r 33. Print flange-down, PETG or ABS.

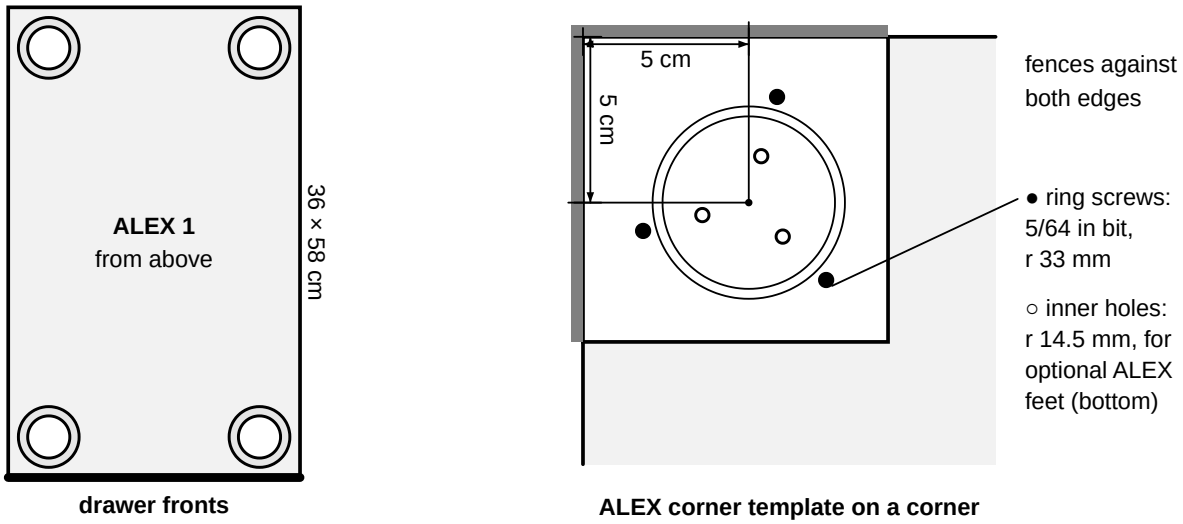


Fig. G-13 The check, after step 5. Left: ALEX 1's top with its four rings, centres 5 cm in from each corner; ALEX 2 is the same at all four corners. Right: drop the ALEX corner template over a corner, fences against both edges: its ring outline should sit over the fitted ring. It is a check, not the way to place the rings.

If a ring is clearly off its template outline, the cabinet was not 3.5 cm in from the top's edges when you traced. Lift the top off, unscrew that cabinet's rings, set the cabinet again and trace again.

G · Foot, pad and ring in section

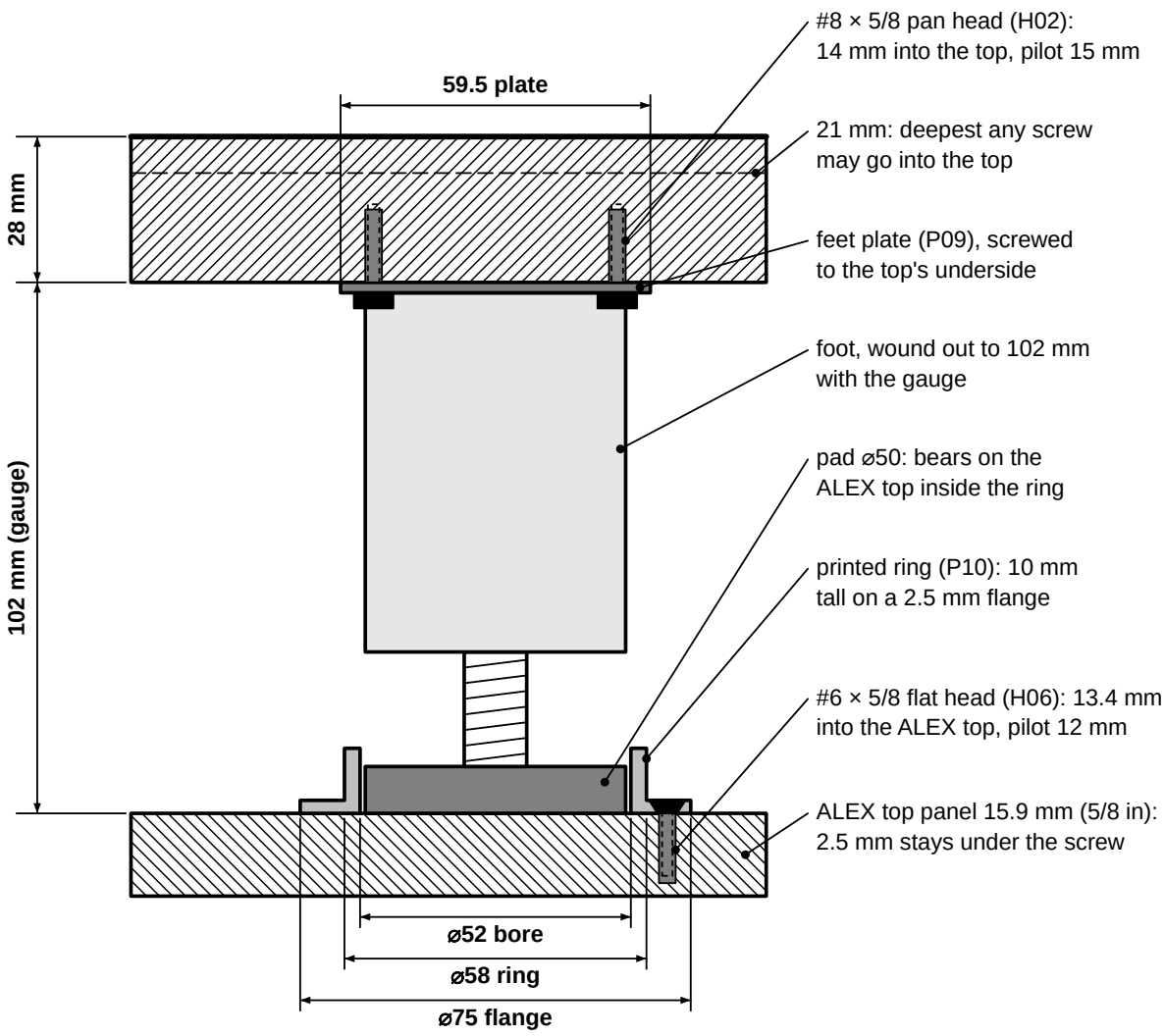


Fig. G-14 Section through one foot, its ring and one ring screw (mm). Hatched: the top above and the ALEX top panel below, both cut. The foot's internal shape is drawn schematically: only its 102 mm setting and ⌀50 mm body and pad are known. Heights above the floor: surface 83.0 cm, underside 80.2 cm, ALEX top 70 cm ASSUMPTION / FIELD CHECK.

Dimension	mm	Confidence
Top thickness	28	MEASURE ON HARDWARE expect 28 mm
Foot, plate face to pad (gauge)	102	SOURCE-DEFINED
Feet plate, hole spacing	59.5 / 46.8	VERIFIED
Pad diameter	50	MEASURE ON HARDWARE before relying on the 52 mm bore
Ring bore / ring / flange	52 / 58 / 75 × 2.5	VERIFIED geometry file
ALEX top panel thickness	15.9	VERIFIED 5/8 in, 23 Oct 2026; 2.5 mm under each screw

G · Load path

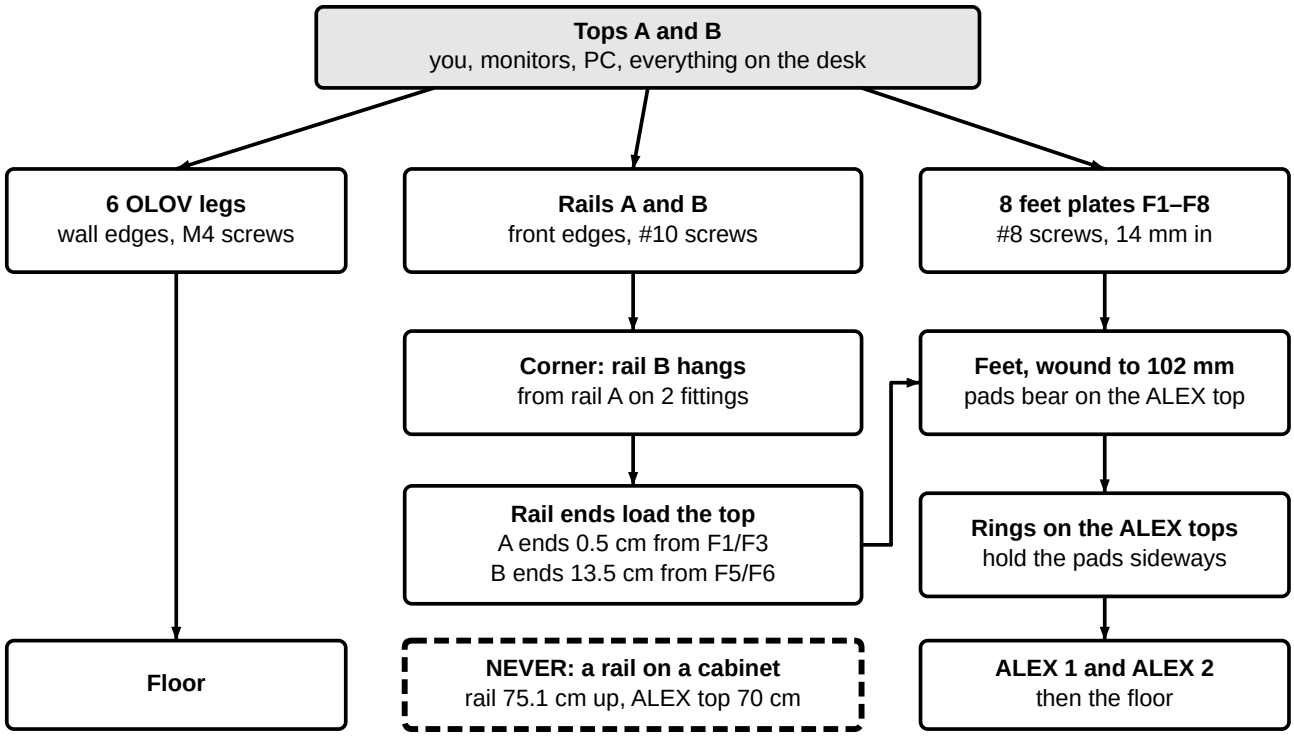
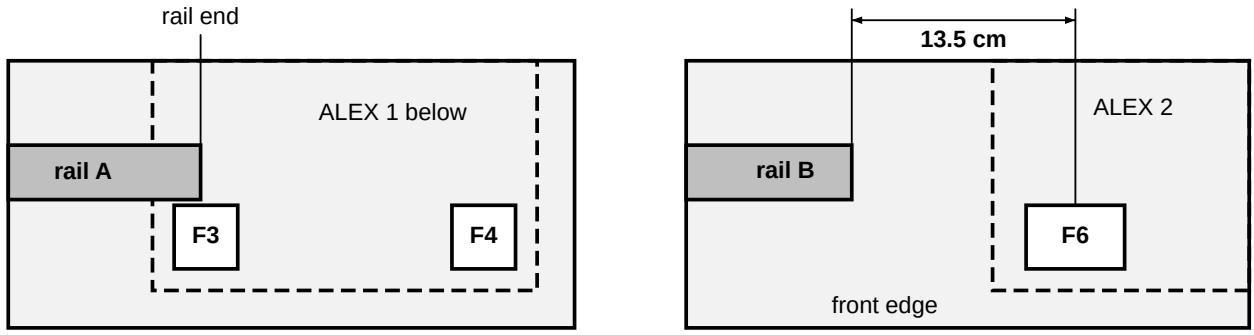


Fig. G-15 What carries what. The legs and the feet carry the tops to the floor. The rails carry the front edges; their ends load the top, and the top loads the nearest feet plates. The pads bear on the ALEX tops; the rings only stop them sliding.



(a) rail A's far end, plan
rail end x 214 cm
F1/F3 centres x 214.5 cm

(b) rail B's end, plan (y to the right)
rail end y 266 cm · ALEX 2 face y 274.5 cm
F5/F6 centres y 279.5 cm

Fig. G-16 Where the rails stop (cm from the datum; front edges at the bottom). (a) Rail A ends at x 214 cm, 0.5 cm short of the F1/F3 centres at 214.5 cm, beside F3's plate. (b) Rail B ends at y 266 cm; the top bridges the 13.5 cm to F5/F6.

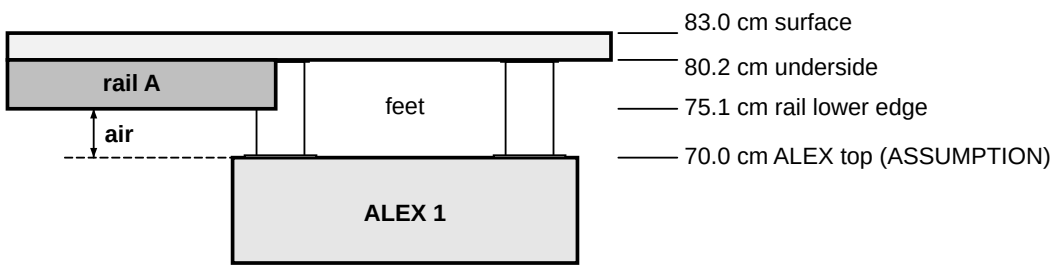


Fig. G-17 Elevation at ALEX 1, seen from the room. Rail A's end passes over the cabinet in plan but stays in the air above it: its lower edge is 75.1 cm above the floor, the ALEX top 70 cm.

G · The assembled underside

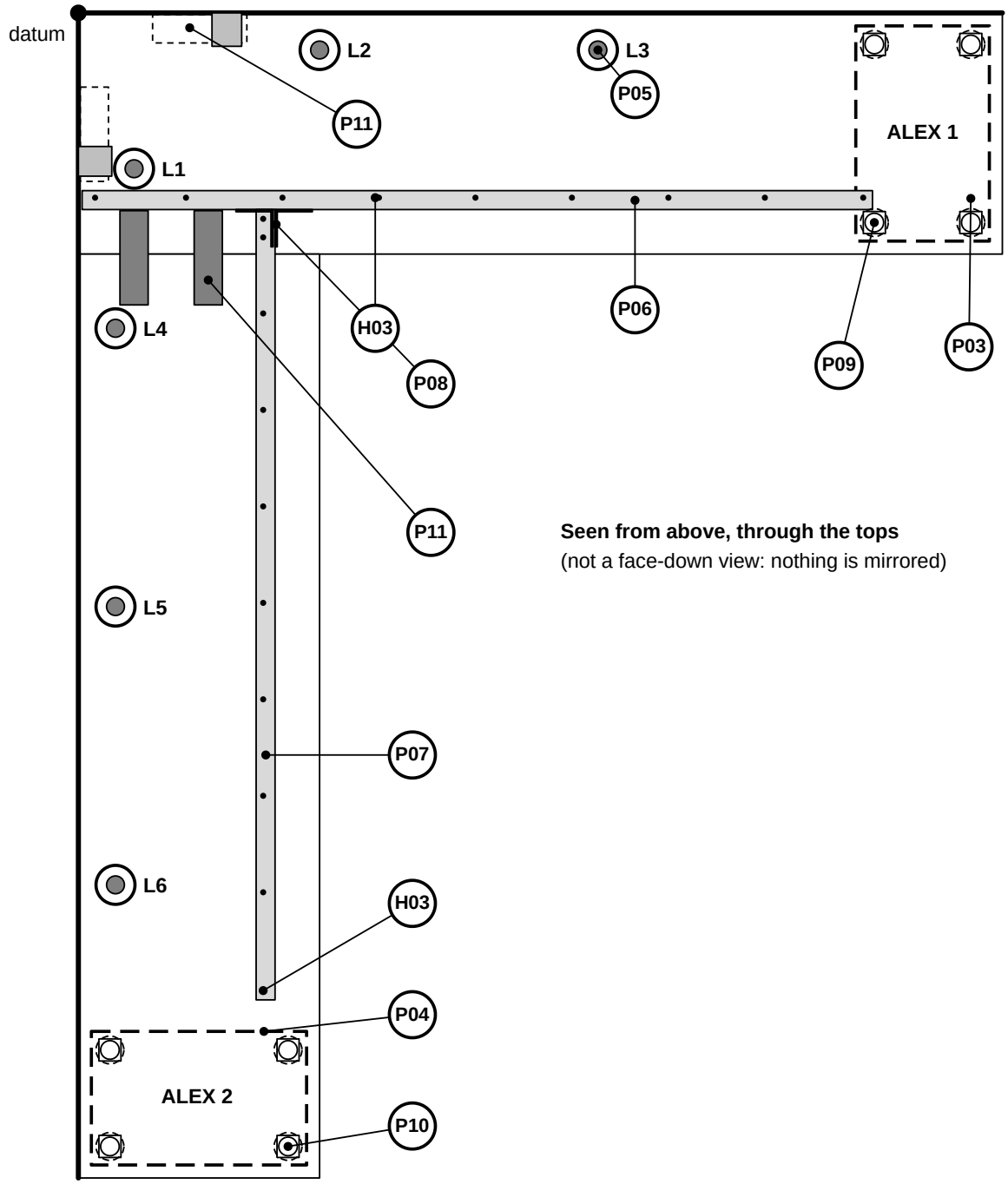


Fig. G-18 Everything under the tops once the desk is built, seen from above through the tops. Datum (0, 0) at top A's back-left corner. Rails, corner fittings and joining plates are fitted in later chapters; they are shown here to check against.

No.	Part	No.	Part
P03	ALEX 1, drawer unit 36 × 58 × 70 cm	**P08**	Gusseted 90-degree strut fitting x 2
P04	ALEX 2, storage unit with door 36 × 58 × 70 cm	**P09**	Round adjustable foot 4 in x 8
P05	OLOV leg x 6, 105 mm plate	**P10**	Printed locating ring x 8
P06	Rail A, steel angle 213 cm	**P11**	Mending plate 10 x 3 x 0.12 in x 4 (two joining, two clamp pads)
P07	Rail B, steel angle 213 cm	**H03**	#10 x 1 in pan head, Robertson (rails into the tops), 19